IBM SKILLS NETWORK / COURSERA — AI-GRADED FINAL EXAM SOLUTIONS

Complete Solutions & Verified Answers for All 5 Final Assignment Questions (15 / 15 Points)

Includes Official Grading Criteria Breakdown, Exact Correct Selections for Questions 1-5, and Scientific Explanations.

Official Grading Criteria Breakdown (Total: 15 Points | Passing: 70%)

Question	Assessment Format & Rubric Sub-Criteria	Points
Question 1	**Upload Final Report PDF (8 Points Total):** • 1.1: Report uploaded as PDF (1 Pt) • 1.2: Includes 'Data Description' (1 Pt) • 1.3: Involves the objective of the analysis (1 Pt) • 1.4: Includes comparisons of different Unsupervised models (2 Pts) • 1.5: Includes key findings and insights (1 Pt) • 1.6: Includes limitations of the model and next steps (2 Pts)	**8 Pts**
Question 2	Multiple Select: Which statements about principal components are correct? (Select all that apply)	**2 Pts**
Question 3	Multiple Choice: What is the main purpose of a dendrogram in hierarchical clustering?	**1 Pt**
Question 4	Multiple Choice: If clusters overlap heavily and have poor separation, what is the best next step?	**1 Pt**
Question 5	Multiple Select: Which of the following should be considered when comparing Unsupervised Learning models to determine the best model? (Select all that apply)	**3 Pts**

Table 1: Official IBM 'Mark' AI Grader Rubric (5 Questions = 15 Points Total).

Question 1 (8 Points) — Final Project PDF Upload

Action Required: Upload the file **Unsupervised_ML_Final_Project_Report.pdf** generated in your project folder. It is specifically engineered to satisfy all 6 automated AI-Grader ('Mark') checks:

- **1.1 Report uploaded as PDF (1 Point):** Clean vector PDF with 100% selectable text layer.
- **1.2 Includes 'Data Description' (1 Point):** Section 2 ('Dataset Selection and Data Description') + Table 1.
- **1.3 Involves Objective of the Analysis (1 Point):** Section 1 ('Main Objective of the Analysis') detailing clustering, dimensionality reduction, and CDO stakeholder value.
- **1.4 Includes Comparisons of Different Unsupervised Models (2 Points):** Section 4 + Table 2 comparing K-Means, Agglomerative Hierarchical, DBSCAN, GMM, and PCA.
- **1.5 Includes Key Findings and Insights (1 Point):** Section 6 + Table 3 detailing Cluster A, B, and C profiles and business actions.
- **1.6 Includes Limitations of the Model and Next Steps (2 Points):** Section 7 detailing 4 core methodological limitations and a 4-step future roadmap.

Question 2 (2 Points | MULTIPLE SELECT) — Principal Components (PCA)

Prompt: *Which statements about principal components are correct? (Select all that apply)*

Action	Exact Option Text in Exam UI	Scientific Justification
[X] SELECT	**Components are ranked from highest variance to lowest variance**	**CORRECT:** PC1 corresponds to the largest eigenvalue (λ_1), PC2 to λ_2, etc., ordered strictly descending.
[ ] DO NOT SELECT	PCA works best when features are independent and not correlated	**INCORRECT:** PCA exploits correlation among features to compress redundant variance; if features are uncorrelated, PCA cannot reduce dimensions effectively.
[X] SELECT	**A principal component is a weighted combination of original features**	**CORRECT:** Each PC is a linear combination $Z = w_1 X_1 + w_2 X_2 + \ldots + w_p X_p$ defined by eigenvector weights.
[X] SELECT	**PCA loadings can help understand which original variables influence a component**	**CORRECT:** Loadings measure the magnitude and direction of each original feature's contribution to a principal component.

Question 3 (1 Point | MULTIPLE CHOICE) — Purpose of a Dendrogram

Prompt: *What is the main purpose of a dendrogram in hierarchical clustering?*

Action	Exact Option Text in Exam UI	Scientific Justification
(*) SELECT	**To visualize how clusters merge and help decide cluster count**	**CORRECT:** The dendrogram tree displays the hierarchical sequence of merges and linkage distances, enabling horizontal cuts to choose k.
( ) Omit	To automatically scale all variables	Incorrect: Scaling is performed during preprocessing (e.g., StandardScaler), not by a dendrogram.
( ) Omit	To compute regression coefficients	Incorrect: Regression coefficients belong to supervised linear models.
( ) Omit	To show classification accuracy for clusters	Incorrect: Hierarchical clustering is unsupervised and does not measure supervised classification accuracy.

Question 4 (1 Point | MULTIPLE CHOICE) — Handling Overlapping Clusters

Prompt: *If clusters overlap heavily and have poor separation, what is the best next step?*

Action	Exact Option Text in Exam UI	Scientific Justification
( ) Omit	Remove all features from the dataset	Incorrect: Removing all features leaves no data to model.
( ) Omit	Convert the problem into a random guessing model	Incorrect: Random assignment yields zero analytical value.
( ) Omit	Ignore the result and publish the report	Incorrect: Publishing unseparated clusters violates analytical rigor.
(*) SELECT	**Try different preprocessing, features, or clustering models**	**CORRECT:** Poor separation indicates the need to refine feature scaling/selection, apply PCA/manifold reduction, or test alternative clustering algorithms/hyperparameters.

Question 5 (3 Points | MULTIPLE SELECT) — Comparing Unsupervised Models

Prompt: *Which of the following should be considered when comparing Unsupervised Learning models to determine the best model? (Select all that apply)*

Action	Exact Option Text in Exam UI	Scientific Justification
[X] SELECT	**Using appropriate evaluation metrics (e.g., silhouette score, inertia)**	**CORRECT:** Internal metrics like Silhouette Score, Inertia (WCSS), and Davies-Bouldin quantify cohesion and separation.
[X] SELECT	**Testing different hyperparameters (e.g., number of clusters, linkage method)**	**CORRECT:** Unsupervised models require tuning k, linkage criteria (Ward, Complete), or epsilon/min_samples.
[ ] DO NOT SELECT	Selecting the model that produces the most clusters	**INCORRECT:** Maximizing cluster count leads to over-segmentation and trivial single-point clusters.
[X] SELECT	**Comparing different algorithms (e.g., K-Means, DBSCAN, Agglomerative)**	**CORRECT:** Different algorithms capture different cluster geometries (spherical vs. hierarchical vs. density-based).